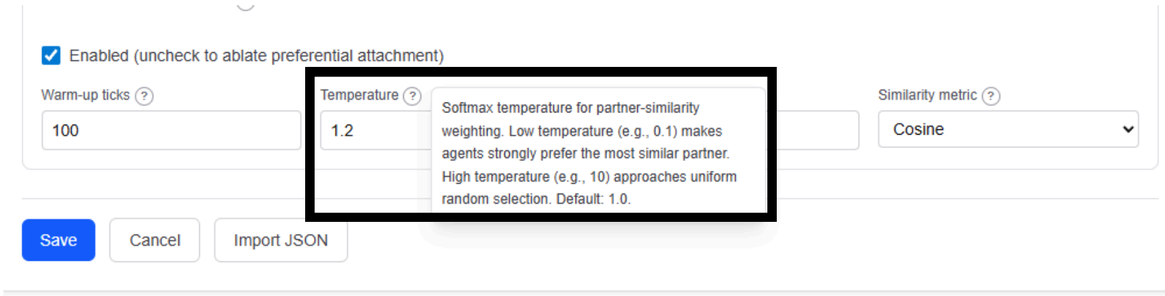
Gaussian Communication Success and agents steps rules

We will measure the communication success (here Probability of Success Ps) by calculating the Euclidean distance of the tokens of agents 1 and agent 2 in World 1 and World 2

E.g. If agent 1 token x has 0.8 weight and agent 2 token u has 0.3, the gap or distance is large enough for the agents do not communicate successfully to match because the tokens are two distant.



In our simulation **Sigma** is the **temperature.**

Sigma is a constant that represents the difficulty metric of a phenomenon.

- **By choosing a large sigma, I increase difficulty to reach communication success (stretching the gap between lexical tokens).**
- **reducing the sigma, I decrease that difficulty (shrinking the gap between the tokens of the agents).**

$$P_s(i,j) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}\right)$$

$$P_s(i,j) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2T^2}\right)$$

- $P_s(i,j)$: The **Probability of Success**, representing the likelihood of effective communication between agent i and agent j within the range $[0,1]$.
- $\mathbf{x}_i, \mathbf{x}_j$: The **Linguistic State Vectors**, containing the numerical weights assigned to specific lexemes by the speaker and listener respectively.
- $\|\mathbf{x}_i - \mathbf{x}_j\|^2$: The **Squared Euclidean Distance**, which quantifies the "Linguistic Gap" or the disparity in intensity of commitment to the vocabulary used.
- σ: The **Kernel Width**, a parameter defining the sensitivity of the success probability to differences in agent states. This is where difficulty is measured by increasing this sensitivity which increase the deviation from the mean at the center or reduce it, by getting closer to the middle point.
- T: The **Social Temperature**, a global simulation parameter that scales the tolerance of the environment; higher T allows for successful communication across larger linguistic distances. We will use the same temperature as in your simulation [configuration] page.

Agents move rule

Agents move after the cosine similarity index reaches between [1;0].

- at [1;0.5] cosine is high, agents sharing this similarity should move one step forward or closer toward each other

Agents move rule

Agents move after the cosine similarity index reaches between [1;0].

- at [0.5;0] cosine is low, agents sharing this similarity should move two steps backward or farther away from each other